Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve flood prediction by leveraging machine learning techniques to analyze environmental parameters and estimate flood risk with high accuracy. The system provides a secure web-based platform where users can register, log in, enter flood-related data, and receive instant predictions. By combining historical flood data with predictive analytics, the application supports early disaster preparedness and informed decision-making for both authorities and the public.

Project Overview	
Objective	Develop an intelligent flood prediction system that uses machine learning algorithms to analyze environmental factors and provide accurate flood risk predictions through an interactive web application.
Scope	The project covers user authentication, flood prediction, prediction history, dashboard analytics, and machine learning integration using Flask and SQLite. It is designed to support disaster management agencies, government authorities, and citizens in making proactive decisions during flood-prone situations.
Problem Statement	
Description	Floods are among the most destructive natural disasters, causing significant damage to life, property, agriculture, and infrastructure. Traditional forecasting methods often struggle to provide accurate

	and timely predictions due to the complexity of environmental factors.
Impact	Delayed or inaccurate flood predictions reduce disaster preparedness and increase economic losses. A reliable machine learning-based prediction system can improve early warning capabilities, reduce response time, and support effective disaster management.
Proposed Solution	
Approach	Build a Flask-based web application integrated with a trained machine learning model to analyze user-provided environmental parameters and predict flood occurrence. The application securely stores prediction records, supports authenticated users, and provides quick prediction results through an intuitive dashboard.
Key Features	• Secure User Registration and Login. • Role-based Administrator and User Access. • Flood Prediction using Machine Learning (XGBoost). • Fast prediction with probability score. • Prediction History stored in SQLite Database. • Dashboard for monitoring predictions. • Responsive web interface. • Future support for live weather API integration and GIS mapping.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU specifications	Intel Core i5 / AMD Ryzen 5 or higher
Memory	RAM required for application and model processing	Minimum 8 GB
Storage	Space for datasets, models, logs, and database	20 GB Available Storage

Software		
Frameworks	Python frameworks	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, XGBoost, Pandas, NumPy, Joblib, Matplotlib, Seaborn
Development Environment	IDE & debugging tools	VSCode, Jupyter Notebook, SQLite CLI, and web developer tools.
Data		
Dataset	Historical flood and weather dataset	Flood Prediction Dataset (CSV format), containing environmental attributes such as rainfall, river water level, humidity, temperature, soil moisture, and flood labels